Student Performance: Statistical Analysis Report

Welch t-test, one-way ANOVA, correlation, multiple regression and chi-square test (APA 7 style)

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1 Summary

This report analyses the public *Student Performance* dataset (Cortez & Silva, 2008; Cortez, 2008): 395 secondary-school mathematics students from two Portuguese schools. Five analyses are shown, each with an assumption check, an APA-style table or figure, and a plain-language interpretation. Final grade (G3) is scored 0 to 20.

Analysis	Statistic	p	Effect size	Result
Welch t-test: G3 by internet access	t(94.22) = -1.99	.050	d = 0.26 (small)	Significant
One-way ANOVA: G3 by study time	F(3, 391) = 1.73	.161	eta2 = .013 (small)	Not significant
Pearson correlation: G1 and G3	r(393) = .80	< .001	r = .80 (large)	Significant
Multiple regression: G3 on study time, absences, failures	F(3, 391) = 20.29	< .001	R2 = .135	Significant
Chi-square: school by internet access	chi2(1) = 7.05	.008	V = .13 (small)	Significant

Note. Effect-size labels follow Cohen's conventions. χ^2 is written chi2, η^2 is eta2 and R^2 is R2 in this table only.

2 Data and descriptive statistics

The file `student-mat.csv` contains 33 variables for 395 students with no missing values. Variables used here are the first-, second- and final-period grades (G1, G2, G3), weekly study time (1 = less than 2 h, 2 = 2 to 5 h, 3 = 5 to 10 h, 4 = more than 10 h), number of past class failures, number of school absences, school (GP or MS) and home internet access.

Table 1

Descriptive Statistics for Study Variables (N = 395)

Variable	M	SD	Min	Max
First-period grade (G1)	10.91	3.32	3	19
Second-period grade (G2)	10.71	3.76	0	19
Final grade (G3)	10.42	4.58	0	20
Weekly study time (1-4)	2.04	0.84	1	4
Past class failures	0.33	0.74	0	3
Absences	5.71	8.00	0	75

3 Welch independent-samples t-test

Research question. Do students with home internet access have different final grades from students without it?

Assumptions. Welch's test does not assume equal variances. Levene's test (median-centred) agreed that variances were similar, *F* = 0.17, *p* = .683. Shapiro-Wilk tests showed departures from normality in both groups (largest *p* < .001), which is expected for bounded integer grades. With 66 and 329 students per group, the t-test is fairly robust, but the result should be read with care.

Table 2

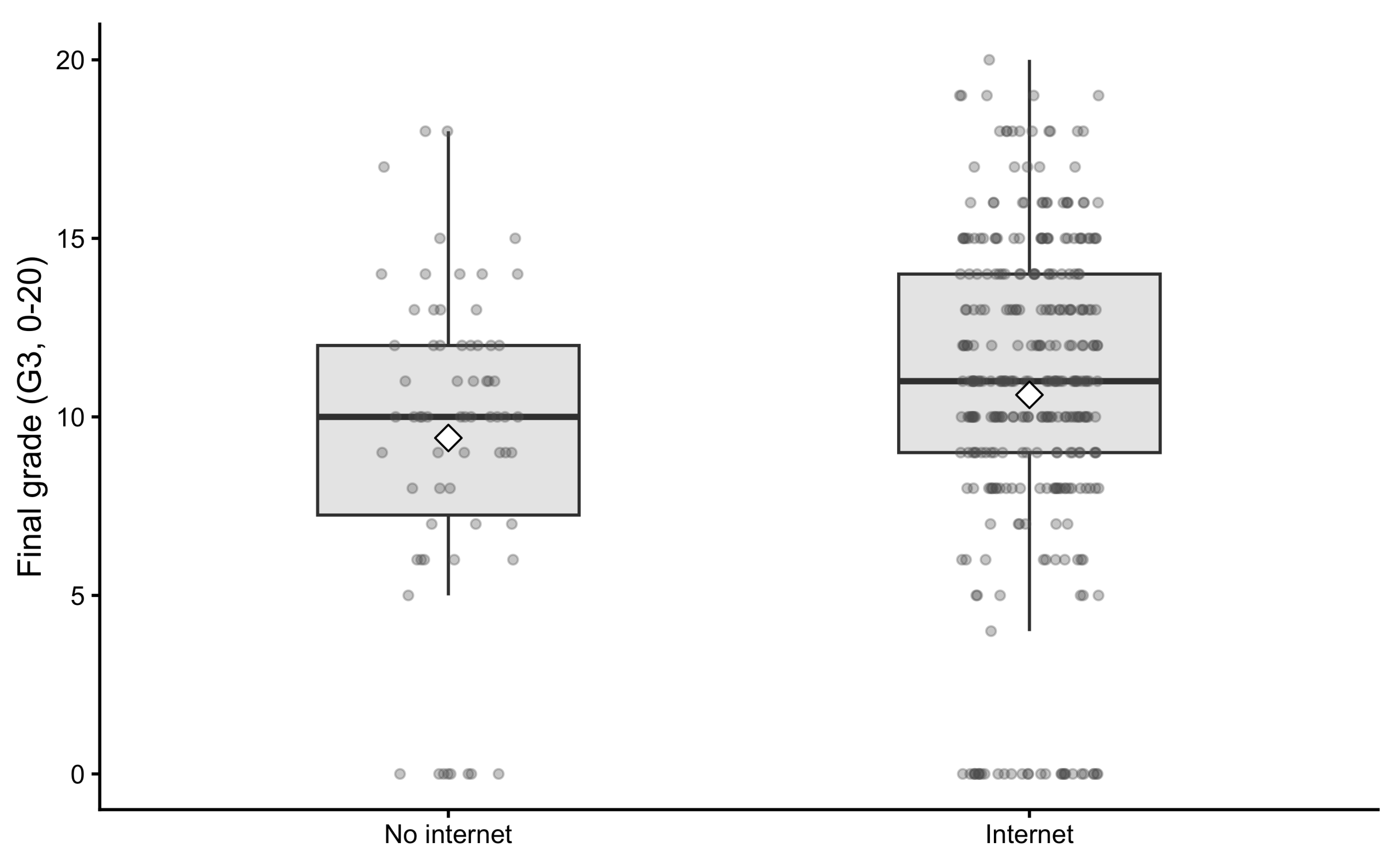
Final Grade (G3) by Home Internet Access

Group	n	M	SD
No internet	66	9.41	4.49
Internet	329	10.62	4.58

Note. Welch test, difference (No internet minus Internet) = -1.21, 95% CI [-2.41, -0.00], *t*(94.22) = -1.99, *p* = .050, Cohen's *d* = 0.26.

Figure 1

Final Grade by Home Internet Access



Note. Diamonds show group means; boxes show medians and quartiles.

Interpretation. Students with internet access (*M* = 10.62, *SD* = 4.58) scored 1.21 points higher than students without it (*M* = 9.41, *SD* = 4.49). The difference was statistically significant, *t*(94.22) = -1.99, *p* = .050, with a small effect (*d* = 0.26). The p value (0.0496 before rounding) sits right on the .05 boundary, and the confidence interval almost touches zero, so this is weak evidence and should be replicated before being relied on. Because the data are observational, internet access may stand in for family resources and should not be read as a cause of higher grades.

4 One-way ANOVA

Research question. Do final grades differ across the four weekly study-time levels?

Assumptions. Levene's test indicated similar variances across groups, *F*(3, 391) = 1.29, *p* = .277. Residuals departed from normality (Shapiro-Wilk *p* < .001), which is common for grade data; the groups are unequal in size (105, 198, 65, 27), so conclusions are tentative.

Table 3

Final Grade (G3) by Weekly Study Time

Study time	n	M	SD	95% CI
<2 h	105	10.05	4.96	[9.09, 11.01]
2-5 h	198	10.17	4.22	[9.58, 10.76]
5-10 h	65	11.40	4.64	[10.25, 12.55]
>10 h	27	11.26	5.28	[9.17, 13.35]

Note. One-way ANOVA: *F*(3, 391) = 1.73, *p* = .161, η^2 = .013.

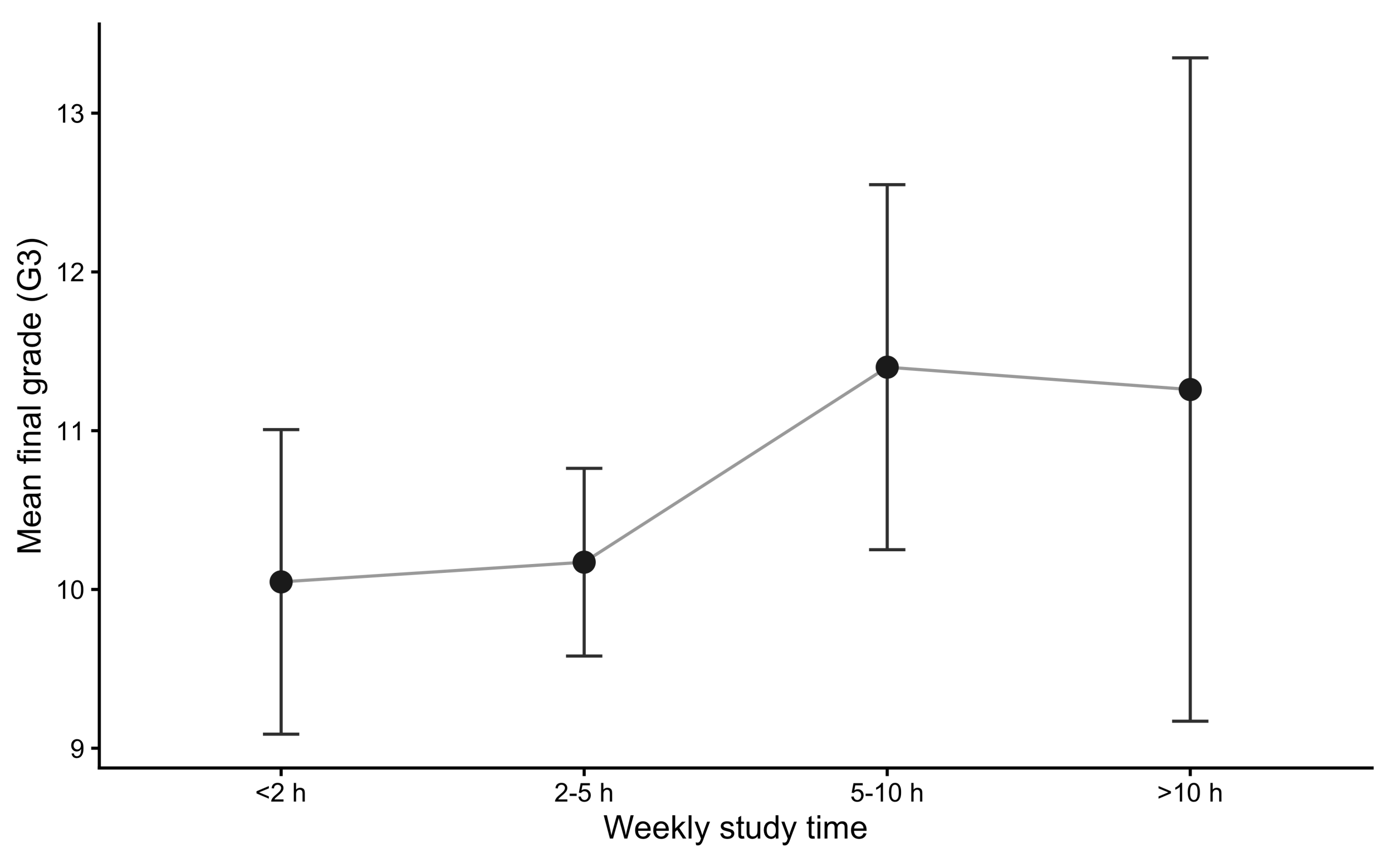
Table 4

Tukey HSD Pairwise Comparisons of Study-Time Groups

	Comparison	Mean difference	95% CI	p (adjusted)
2-5 h-<2 h	2-5 h-<2 h	0.12	[-1.30, 1.55]	.996
5-10 h-<2 h	5-10 h-<2 h	1.35	[-0.51, 3.21]	.240
>10 h-<2 h	>10 h-<2 h	1.21	[-1.33, 3.76]	.609
5-10 h-2-5 h	5-10 h-2-5 h	1.23	[-0.46, 2.91]	.238
>10 h-2-5 h	>10 h-2-5 h	1.09	[-1.33, 3.51]	.652
>10 h-5-10 h	>10 h-5-10 h	-0.14	[-2.84, 2.56]	.999

Figure 2

Mean Final Grade by Weekly Study Time



Note. Error bars show 95% confidence intervals.

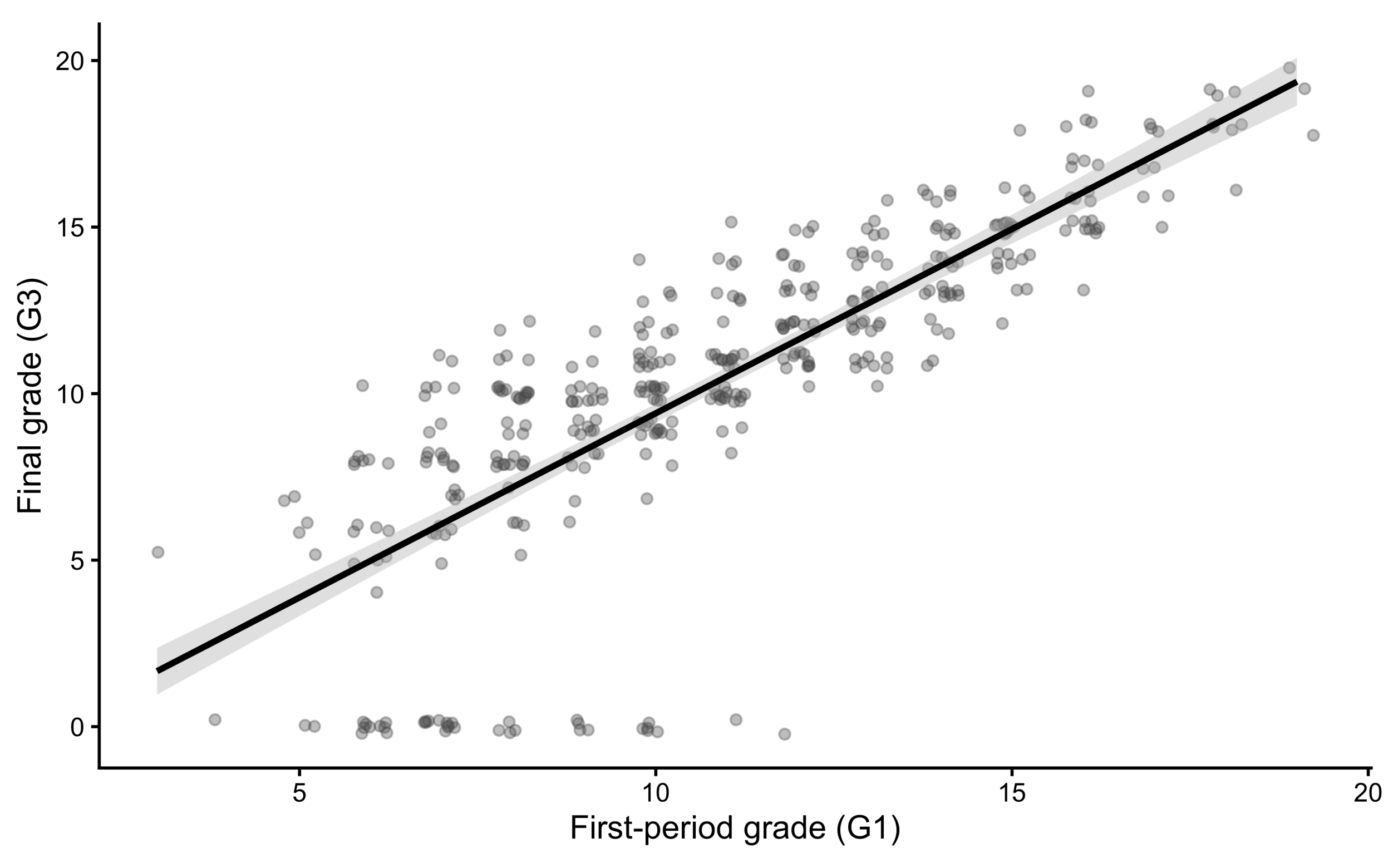
Interpretation. Mean grades rose from 10.05 (less than 2 h) to 11.40 at the highest-scoring study-time group, but the overall effect of study time was not statistically significant, *F*(3, 391) = 1.73, *p* = .161, η^2 = .013 (small effect). No pairwise comparison reached significance after Tukey adjustment. Self-reported study time explains very little of the variation in final grade in this sample.

5 Pearson correlation

Research question. How strongly are first-period (G1) and final (G3) grades related?

Figure 3

First-Period Grade and Final Grade



Note. Points are jittered slightly because grades are whole numbers. The line is the least-squares fit with a 95% confidence band.

Result. G1 and G3 were strongly and positively correlated, *r*(393) = .80, 95% CI [.76, .83], *p* < .001. G1 shares about 64.2% of its variance with G3 (r^2 = .64). Students who start well tend to finish well, which is useful for early-warning systems but says nothing about why.

6 Multiple linear regression

Research question. How well do weekly study time, absences and past failures together predict the final grade?

Table 5

Regression of Final Grade (G3) on Study Time, Absences and Past Failures

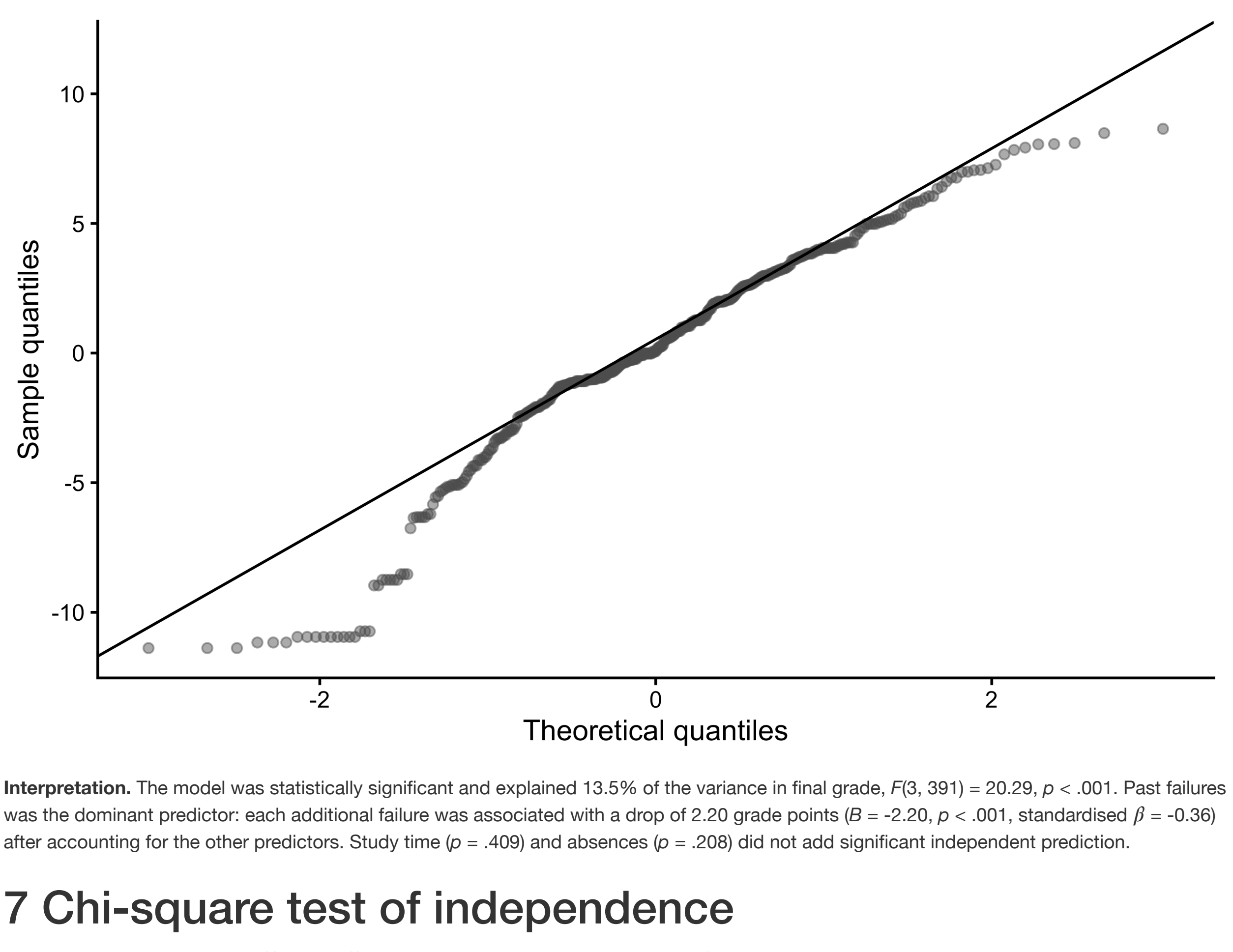
	Predictor	B	SE	Std. beta	t	p	95% CI
(Intercept)	Intercept	10.52	0.62		16.90	< .001	[9.29, 11.74]
studytime_n	Weekly study time (1-4)	0.22	0.26	0.04	0.83	.409	[-0.30, 0.73]
absences	Absences	0.03	0.03	0.06	1.26	.208	[-0.02, 0.09]
failures	Past class failures	-2.20	0.29	-0.36	-7.47	< .001	[-2.78, -1.62]

Note. N = 395. R^2 = .135, adjusted R^2 = .128, *F*(3, 391) = 20.29, *p* < .001. Residual *SE* = 4.28.

Assumptions. The largest Cook's distance was 0.084 (well below 1), so no single student drives the result. Residuals deviated from normality (Shapiro-Wilk *p* < .001, see Figure 4), so standard errors and *p* values should be read as approximate.

Figure 4

Normal Q-Q Plot of Regression Residuals



Interpretation. The model was statistically significant and explained 13.5% of the variance in final grade, *F*(3, 391) = 20.29, *p* < .001. Past failures was the dominant predictor: each additional failure was associated with a drop of 2.20 grade points (*B* = -2.20, *p* < .001, standardised β = -0.36) after accounting for the other predictors. Study time (*p* = .409) and absences (*p* = .208) did not add significant independent prediction.

7 Chi-square test of independence

Research question. Is school (GP vs. MS) associated with having internet at home?

Table 6

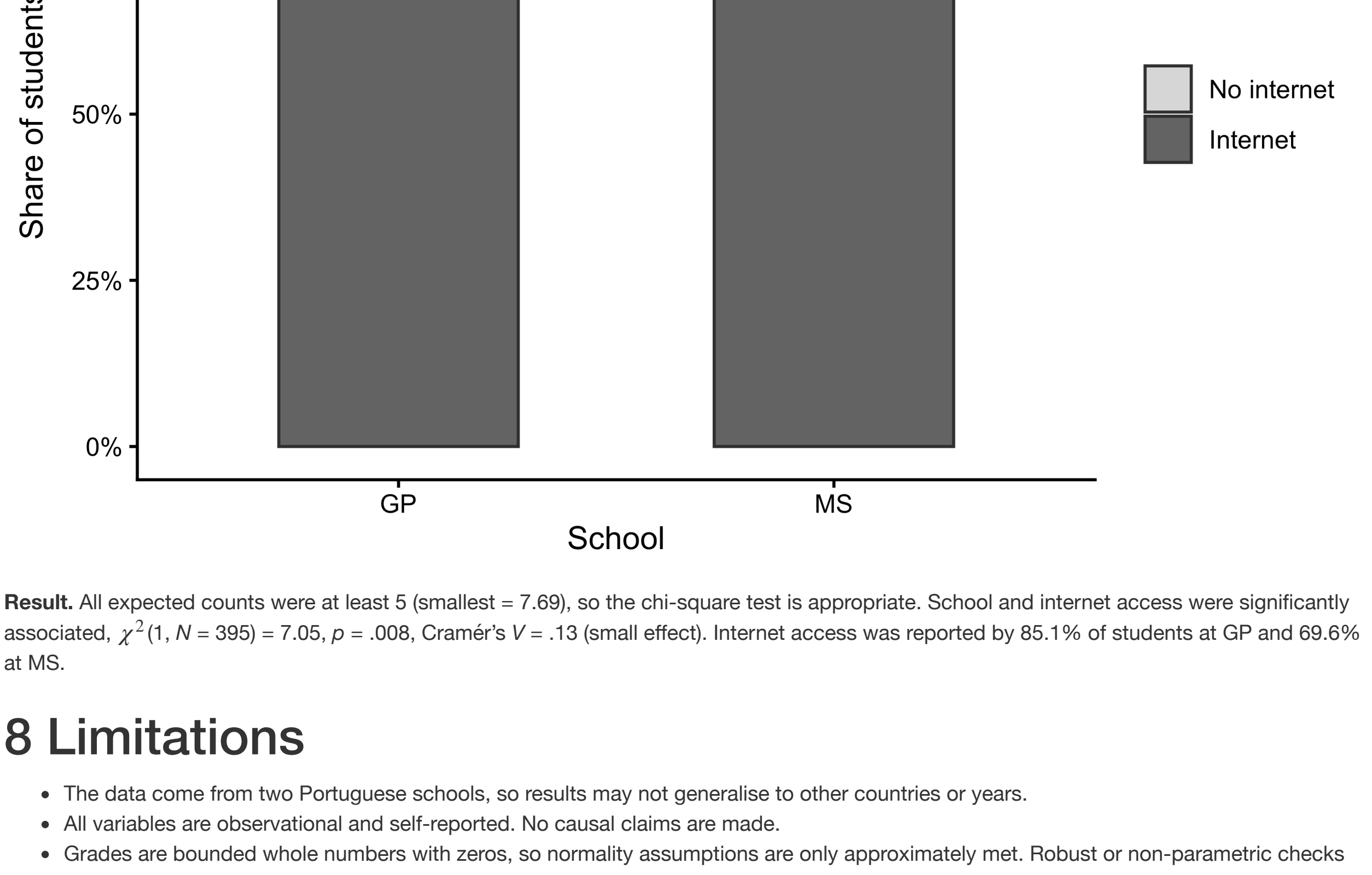
Home Internet Access by School

School	No internet, n (%)	Internet, n (%)
GP	52 (14.9%)	297 (85.1%)
MS	14 (30.4%)	32 (69.6%)

Note. Percentages are row percentages.

Figure 5

Proportion of Students With Home Internet Access, by School



Result. All expected counts were at least 5 (smallest = 7.69), so the chi-square test is appropriate. School and internet access were significantly associated, χ^2(1, *N* = 395) = 7.05, *p* = .008, Cramér's *V* = .13 (small effect). Internet access was reported by 85.1% of students at GP and 69.6% at MS.

8 Limitations

- The data come from two Portuguese schools, so results may not generalise to other countries or years.
- All variables are observational and self-reported. No causal claims are made.
- Grades are bounded whole numbers with zeros, so normality assumptions are only approximately met. Robust or non-parametric checks would be added in a full thesis analysis.
- Five tests were run for demonstration. In a real study, multiple comparisons and a pre-specified analysis plan would be reported.

9 Reproducibility

Check	Matches r-ground-truth.json
Welch t	TRUE
ANOVA F	TRUE
Pearson r	TRUE
Regression R2	TRUE
Chi-square	TRUE

Analysis: R version 4.6.2 (2026-10-29); ggplot2 4.0.3. All code, the cleaned data file and the verification file are available on request and in the project repository.

10 References

Cortez, P. (2008). *Student performance* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

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Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag.